

Leveraging Data Analytics and Behavioural Analysis for Improved Testimony and Incident Correlation

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Project Proposal Report

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Declaration

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Statement of the supervisor,

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

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Abstract

Behavioral and emotional analysis is a critical aspect of criminal investigations, particularly during suspect interrogations, where physiological and emotional cues can provide key insights into deception and stress levels. Traditional interrogation methods rely heavily on subjective human judgment, which is prone to bias and inconsistencies. This research proposes a multi-modal system that integrates facial emotion recognition, IoT-based heart rate monitoring, and voice stress analysis to provide real-time behavioral insights.

The system leverages computer vision techniques to analyze facial expressions and detect emotions such as fear, anxiety, or stress. IoT sensors are used to continuously monitor heart rate variability, capturing physiological responses that may indicate deception or nervousness. Additionally, speech analysis algorithms assess changes in voice patterns, including variations in tone, pitch, and speech rate, to identify vocal stress indicators. These three modalities work in combination to deliver objective, data-driven assessments during interrogations.

A real-time feedback mechanism highlights irregularities in emotional and physiological responses, assisting investigators in evaluating suspect testimonies more effectively. Unlike conventional approaches, this AI-driven system reduces human error, improves investigative accuracy, and enhances deception detection by offering a quantifiable and automated method for behavioral assessment. By integrating IoT and artificial intelligence technologies, this research aims to streamline interrogation processes, provide actionable insights, and support law enforcement in making more informed decisions during investigations.

Keywords – Behavioral Analysis, Emotional Detection, Facial Emotion Recognition, Heart Rate Variability, Voice Stress Analysis, Machine Learning, Computer Vision, Criminal Investigations, Suspect Interrogation

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List of Abbreviations

Abbreviation	Description
AI	Artificial Intelligence
ML	Machine Learning
IDE	Integrated Development Environment
WCAG	Web Content Accessibility Guidelines

1. Introduction

1.1. Background

The role of artificial intelligence (AI) in criminal justice has become increasingly prominent, as AI-based systems help streamline criminal investigations and improve decision-making. In particular, AI applications in deception detection, emotion recognition, and physiological analysis during interrogations are gaining traction in law enforcement and forensic investigations [1] [2]. Modern technologies, such as facial emotion recognition, voice stress analysis, and heart rate monitoring, are showing promise in enhancing the accuracy of behavioral assessments during suspect interrogations. By analyzing emotions and physiological signals, AI can provide valuable insights into a suspect's emotional state, which could help investigators gauge the veracity of testimonies and detect signs of deception.

1.2 Research Gap

While there is substantial research on individual technologies used in behavioral and emotional analysis, there is a notable gap in multi-modal systems that integrate multiple data sources for comprehensive analysis. Research A [3] and Research B [4] explore facial emotion recognition and voice analysis separately, but they do not combine these insights with physiological signals like heart rate monitoring. Additionally, while Research C [5] and Research D [6] focus on stress detection through speech patterns, they do not integrate other modalities like facial analysis or real-time feedback to investigators. Furthermore, current systems often do not provide real-time, actionable feedback for investigators during the interrogation process, limiting their effectiveness. Research E [7] presents a multi-modal sentiment analysis framework but falls short in real-time behavioral assessments that combine emotional and physiological states.

Feature	Research A	Research B	Research C	Research D	Research E	Proposed System
Facial Emotion Recognition	✓	✓	✗	✗	✓	✓

Voice Analysis	✓	✓	✓	✓	✓	✓
Heart Rate Monitoring	X	X	X	X	X	✓
Stress Detection	X	X	✓	✓	✓	✓
Real-Time Feedback for Investigators	X	X	X	X	X	✓
Multi-Modal Integration	X	X	X	X	✓	✓
Emotional & Physiological State Assessment	X	X	X	X	X	✓

Table 1 Research Gap

1.3 Research Problem

The fundamental research problem addresses the imperative need for an integrated, real-time multi-modal system capable of analyzing emotional and physiological behavior during criminal interrogations. As evidenced by Research A and Research E, current investigative procedures predominantly rely on human interpretation of behavioral cues, introducing potential subjective bias and inconsistency. The research challenges encompass:

1. Development of simultaneous processing algorithms for facial, cardiovascular, and vocal data streams in real-time environments
2. Implementation of correlation mechanisms for multiple behavioral indicators to enhance emotional state assessment accuracy
3. Creation of instantaneous feedback systems for investigator support during active interrogations
4. Establishment of robust accuracy metrics across diverse interview conditions and subject demographics

This research endeavors to address these challenges through the development of a comprehensive multi-modal analysis system, integrating advanced computer vision, physiological monitoring, and audio processing technologies, thereby enhancing the objectivity and efficacy of criminal interrogations.

Research Question

- How can combining facial emotion recognition, voice analysis, and heart rate monitoring improve behavioral analysis in criminal interrogations?
- What role does real-time feedback on emotional and physiological data play in enhancing investigator decision-making during interrogations?
- How does integrating multiple data sources (facial, voice, and heart rate) improve stress detection accuracy compared to traditional methods?

2. Objective

2.1. Main Objectives

To develop a multi-modal system for real-time analysis of emotional and physiological behavior during suspect interrogations, integrating facial emotion detection, voice analysis, and heart rate monitoring to provide actionable insights for investigators.

2.2. Sub Objectives

Facial Emotion Detection: Develop a system that uses computer vision techniques, specifically Convolutional Neural Networks (CNNs), to analyze video footage from interrogations and detect key emotions such as fear, anger, and stress.

Heart Rate Monitoring Integration: Integrate heart rate data collected from IoT sensors with other behavioral analysis tools to detect physiological responses indicative of emotional distress or potential deception.

Audio Analysis for Deception Detection: Implement audio analysis algorithms to assess voice patterns, such as pitch, tone, and speed, in order to detect signs of nervousness, stress, or dishonesty.

Real-Time Feedback: Provide real-time feedback to investigators during interrogations, highlighting emotional and physiological irregularities that may suggest deception or emotional distress, enhancing decision-making during the interview process.

3. Methodology

3.1. System Overview

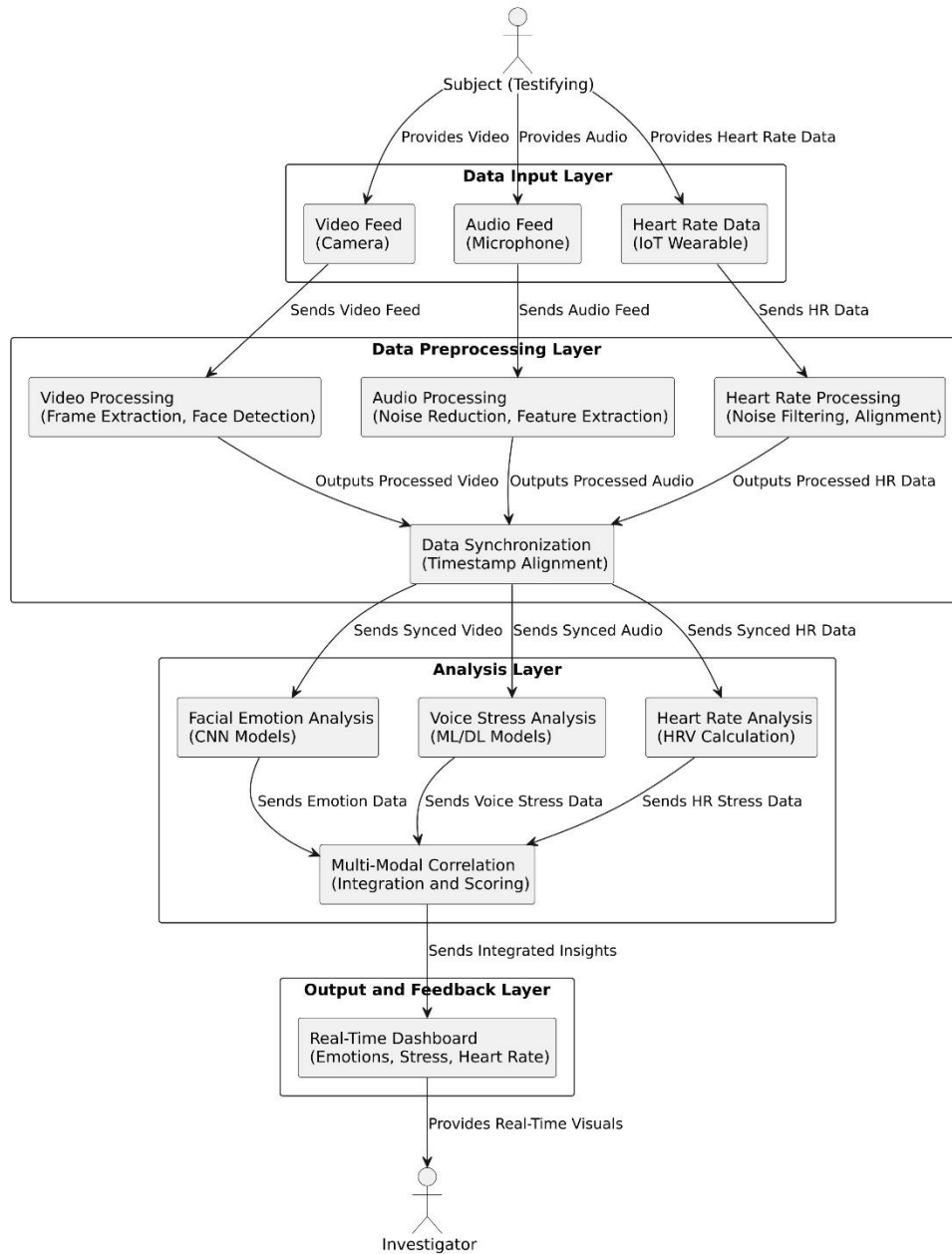


Figure 1 System Overview

The framework begins with the Data Input Layer, where the subject, during testimony, provides three primary forms of raw data: video, audio, and heart rate data. The video is captured through a camera, focusing on the subject's facial expressions and physical movements. The audio feed, recorded via a microphone, provides detailed input on the subject's speech, tone, and vocal nuances. Simultaneously, physiological data is gathered through an IoT wearable device, which

monitors the subject's heart rate. This combination of data inputs lays the foundation for further analysis by providing a holistic view of both external behavior and internal physiological states.

In the Data Preprocessing Layer, each type of data undergoes specialized preprocessing to enhance its quality and extract meaningful features. The video feed is processed to extract frames and perform face detection, isolating critical visual features for further analysis. Audio data is subjected to noise reduction and feature extraction to filter out irrelevant background sounds and focus on the subject's vocal patterns. Similarly, the heart rate data is processed through noise filtering and alignment techniques to ensure accuracy and synchronization with other inputs. These refined data streams are then aligned through a timestamp-based synchronization process to create a cohesive dataset where video, audio, and heart rate data are temporally matched.

Once the data is synchronized, it is passed to the Analysis Layer, where advanced computational models analyze each modality. Facial emotion analysis is performed using convolutional neural networks (CNNs) to detect and interpret the subject's emotional state based on facial expressions. Voice stress analysis leverages machine learning or deep learning models to assess vocal stress patterns and detect signs of anxiety or nervousness. Meanwhile, heart rate variability (HRV) is calculated from the heart rate data to measure stress levels from a physiological perspective. These independent analyses produce emotion data, voice stress data, and heart rate stress data.

The outputs from the analysis stage are integrated in the Multi-Modal Correlation step, which combines the insights from facial expressions, vocal stress, and physiological stress. This integration allows for a comprehensive understanding of the subject's emotional and stress responses by assigning scores and identifying correlations between the modalities. The resulting integrated insights are then sent to the Output and Feedback Layer.

Finally, in the Output and Feedback Layer, the combined insights are visualized on a real-time dashboard. This dashboard provides investigators with a detailed, live overview of the subject's emotions, stress levels, and heart rate data, enabling quick and informed decisions. The visualizations ensure that investigators can easily interpret the subject's state during testimony, facilitating a more nuanced and objective evaluation process.

4. Project Requirements

4.1 Functional & Non-Functional Requirements

Functional Requirements	Non Functional Requirements
Real-Time Data Capture: The system must continuously collect and process video, heart rate, and audio data in real-time.	Scalability: The system should be able to handle multiple interrogation sessions simultaneously, with minimal latency in data processing.
Facial Emotion Recognition: The system should detect and classify emotions (fear, anger, stress, discomfort) from facial expressions with high accuracy.	Performance: The system must process video, audio, and heart rate data with low latency (real-time feedback within seconds of data input).
Heart Rate Analysis: The system should be able to detect abnormalities in heart rate, including elevated rates or irregular HRV indicative of emotional distress or deception.	Usability: The interface should be intuitive and easy to navigate for investigators, providing clear alerts and analysis without overwhelming them.
Voice Analysis: The system should analyze voice patterns, detecting signs of nervousness, hesitation, or contradictions that suggest dishonesty.	Security: All data (video, audio, heart rate) must be encrypted during transmission and stored securely to maintain the privacy of suspects.
Real-Time Feedback: The system should provide visual feedback to the investigator, including color-coded alerts and timestamps highlighting potential emotional or physiological irregularities.	Reliability: The system should be reliable, with minimal downtime, ensuring that the analysis can be conducted during the interrogation without interruption.

Table 2 Functional and Non-functional Requirements

4.2 User Requirements

- **Real-Time Feedback for Investigators:** The system must provide investigators with real-time insights into the emotional and physiological states of suspects during interrogations.
- **Easy-to-Use Interface:** Investigators should have access to an intuitive and user-friendly interface to monitor emotional and physiological data during the interrogation without disrupting the flow of the process.
- **Emotion and Stress Detection:** The system must reliably detect emotions (e.g., anger, fear) and stress signals (e.g., increased heart rate, voice pitch variation) from the suspect.

- **Accurate Data Display:** Emotional and physiological data must be presented clearly, with real-time visualizations (e.g., graphs, charts) for easy interpretation by investigators.
- **Customizable Alerts:** The system should allow investigators to set thresholds for emotional and physiological data (e.g., heart rate, speech rate) to trigger alerts for further attention.
- **Confidentiality and Security:** Sensitive data, including emotional and physiological information, must be stored securely with proper encryption to ensure privacy and confidentiality.
- **Multi-Modal Integration:** The system should combine data from multiple sensors (e.g., facial recognition, voice analysis, heart rate monitoring) to provide a comprehensive understanding of the suspect's behavior.

4.3 Software Requirements

Hardware:

- **IoT Heart Rate Sensors:** Capable of continuously monitoring the heart rate of the suspect during the interrogation.
- **High-Resolution Cameras:** Required for capturing facial expressions and feeding data into the emotion detection system.
- **Microphone/Audio Capture Devices:** For collecting audio data during the interrogation to analyze voice patterns.

Software:

- **Emotion Recognition Software:** Use of deep learning algorithms (e.g., CNNs) for facial emotion detection.
- **Audio Analysis Tools:** Speech processing software capable of analyzing voice features like pitch, speed, and tone.
- **Heart Rate Monitoring Integration:** Software to process data from IoT sensors and integrate it with other data streams.
- **Real-Time Feedback Dashboard:** A graphical user interface (GUI) that presents all the data (facial emotion, voice patterns, heart rate) and alerts the investigator in real-time.

4.4 Work Breakdown Structure

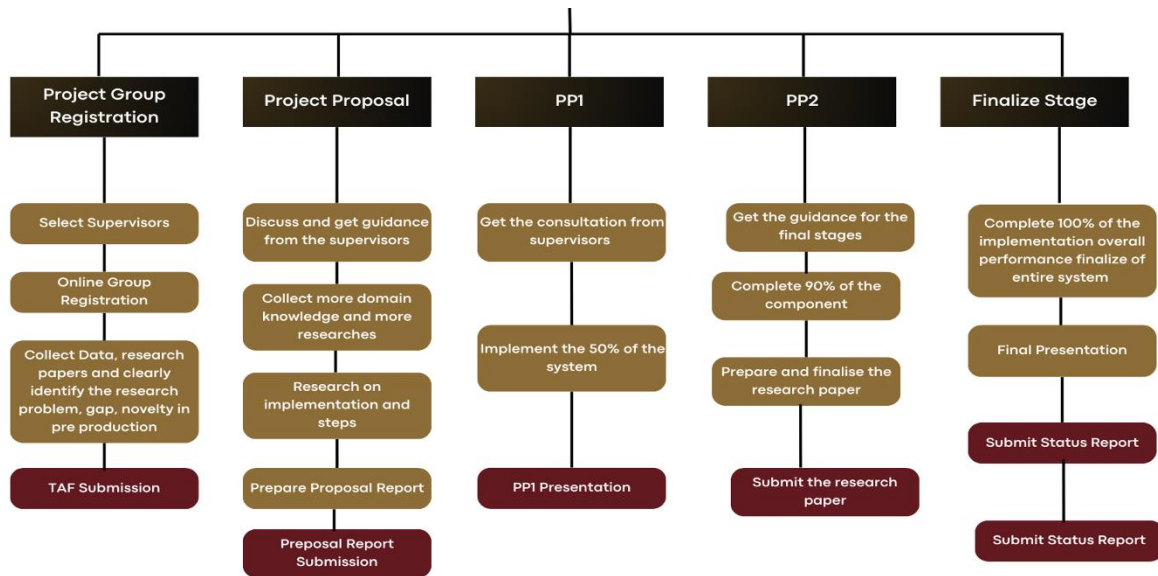


Figure 2 Work Breakdown Structure

4.5 Gantt Chart

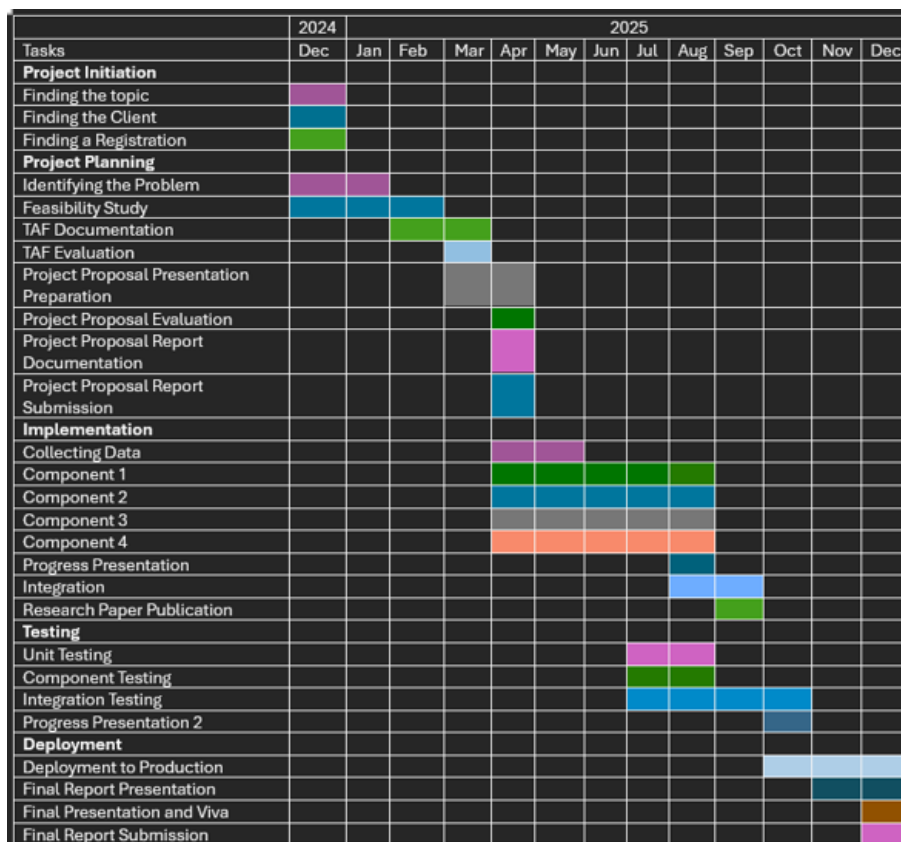


Figure 3 Gantt Chart

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